

Assignment 1 - ABKN05

LCC of Energy Measures for Rönnekyrkan

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Introduction & Objectives

What is LCC?

Life-Cycle Costing evaluates the total cost of a measure over its entire life, including investment, operation, and maintenance.

Objective: *To propose and analyze two energy-saving measures for the Rönnekyrkan church in Ängelholm from an economic perspective using LCC.*

The Two Measures used:

- Hot Water System: Reducing consumption with efficient fixtures.
- Front Door: Reducing heat loss through better doors.

Key Assumptions

Occupants: 25 persons

Labour Cost: 40%

Hot water: 40% of general water consumption

Ground Temperature: 8°C

Water max. heating temperature: 55°C

Heating price (SEK/kWh) - 1.38

Electricity price (SEK/kWh) - 1.60

r_n (nominal interest rate) - 2%

r_f (inflation) - 3%

r_{ci} (nominal price increase) - 2.5%

r_{pi} (real price increase) - 0.49%

r_r (real interest rate) - 0.97%

r_{di} (discount interest rate) - 0.49%

Water price (SEK/Ltr) - 0.03

Measure 1: Hot Water System

Problems: Identified hot water consumption due to old fixtures.

Solution: Install new, efficient faucets and taps to reduce water flow.

(Base Case)



Option 1 (replace fixtures)



(Base Case)

Option 2 (fixtures+Electric water heater)



Measure 1: Hot Water System

Result: Consumption: Both options reduce water use by 73% (from ~13,000L to ~3,300L annually)

(Base Case)

Option 1 (replace fixtures)

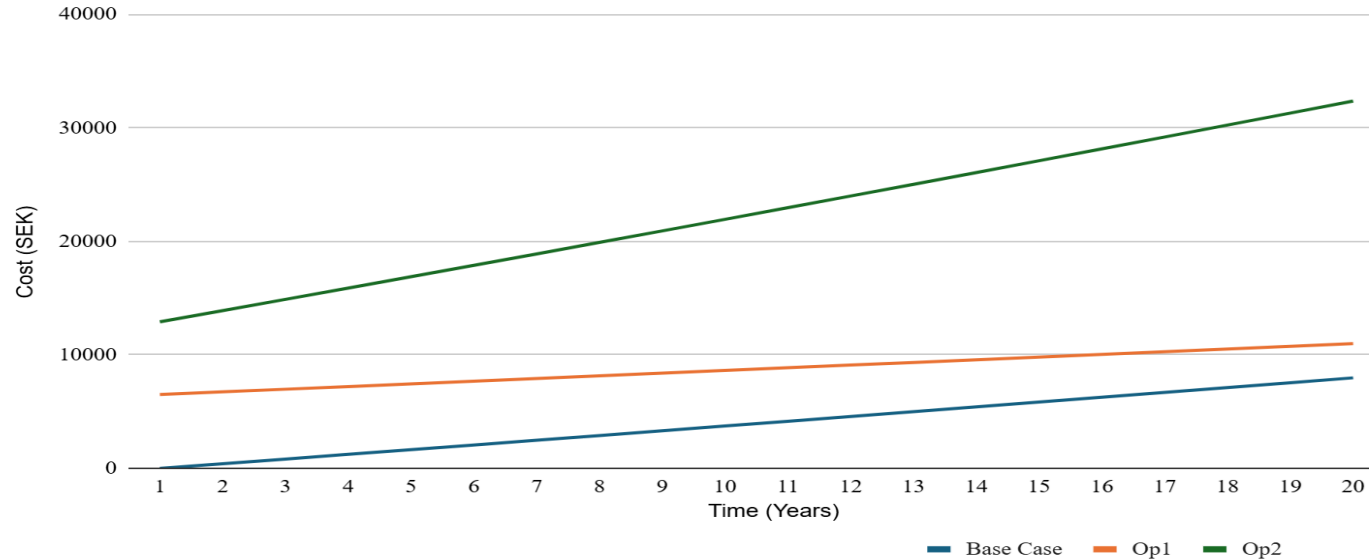
Option 2 (fixtures+heater)

- | | | |
|---------------------|-------------------------|--------------------------|
| ● Investment: 0 SEK | ● Investment: 6,500 SEK | ● Investment: 12,900 SEK |
| ● LCC: 13,701 SEK | ● LCC: 10,772 SEK | ● LCC: 39,209 SEK |

Conclusion: Option 1 is the best available option, saving water and money in the long-term. Option 2 is not economically viable due to high electricity costs.

Measure 1: Hot Water System

LCC analysis of Hot Water Supply



Op1 compared to the Base Case is approximately 34.2 years, which exceeds the analysis period, while Op2 does not achieve payback.

Measure 2: Front Door

- The existing door is old, can be left open for a long period of time

Door 1 (Base Case)



Door 2 (Door Closer)



Door 3 (Automatic Doors)



<https://www.amazon.se/-en/Automatic-Operation-Electric-Shutter-Commercial/dp/B0D76C9RQR>

<https://dtservicesltd.co.uk/automatic-doors/>

Measure 2: Front Door

- The existing door is old, can be left open for a long period of time

Door 1 (Base Case)

- High U-Value
(2.5 W/m²K)
- A long opening
time (3600 s per
day)

Door 2 (Door Closer)

- High U-Value
(2.5 W/m²K)
- A shorter opening
time (1500 s per
day)

Door 3 (Automatic Doors)

- Low U-Value
(1.4 W/m²K)
- The shortest
opening time (750 s
per day)

Measure 2: Front Door

LCC analysis considering:

- Heating losses
 - Conduction
 - Losses from opening
- Maintenance costs
- Initial investment (purchase and workmanship)
- Operational costs (electricity for the equipment)

$$E_{\text{conduction}} = U \times A \times Gh/1000$$

$$Q = \rho \times q \times cp \times (T_i - T_o)$$

$$E_{\text{opening}} = Q \times t$$

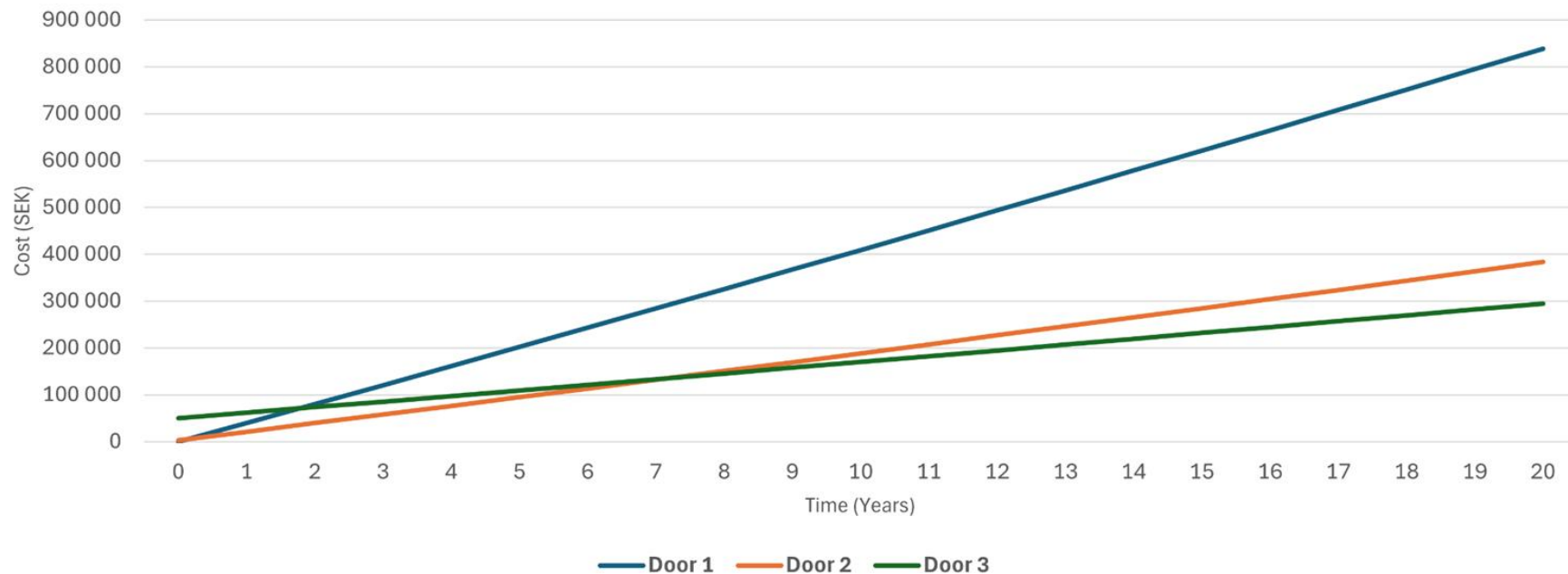
$$q = C_d \frac{B}{3} \sqrt{g' \cdot H^3}$$

$$g' = 2 \cdot g \frac{T_i - T_o}{T_i + T_o}$$

$$C_d = 0.4 + 0.0045 |T_i - T_o|$$

Measure 2: Front Door

LCC with increased heating prices



Thank you for your attention!

Any Questions?